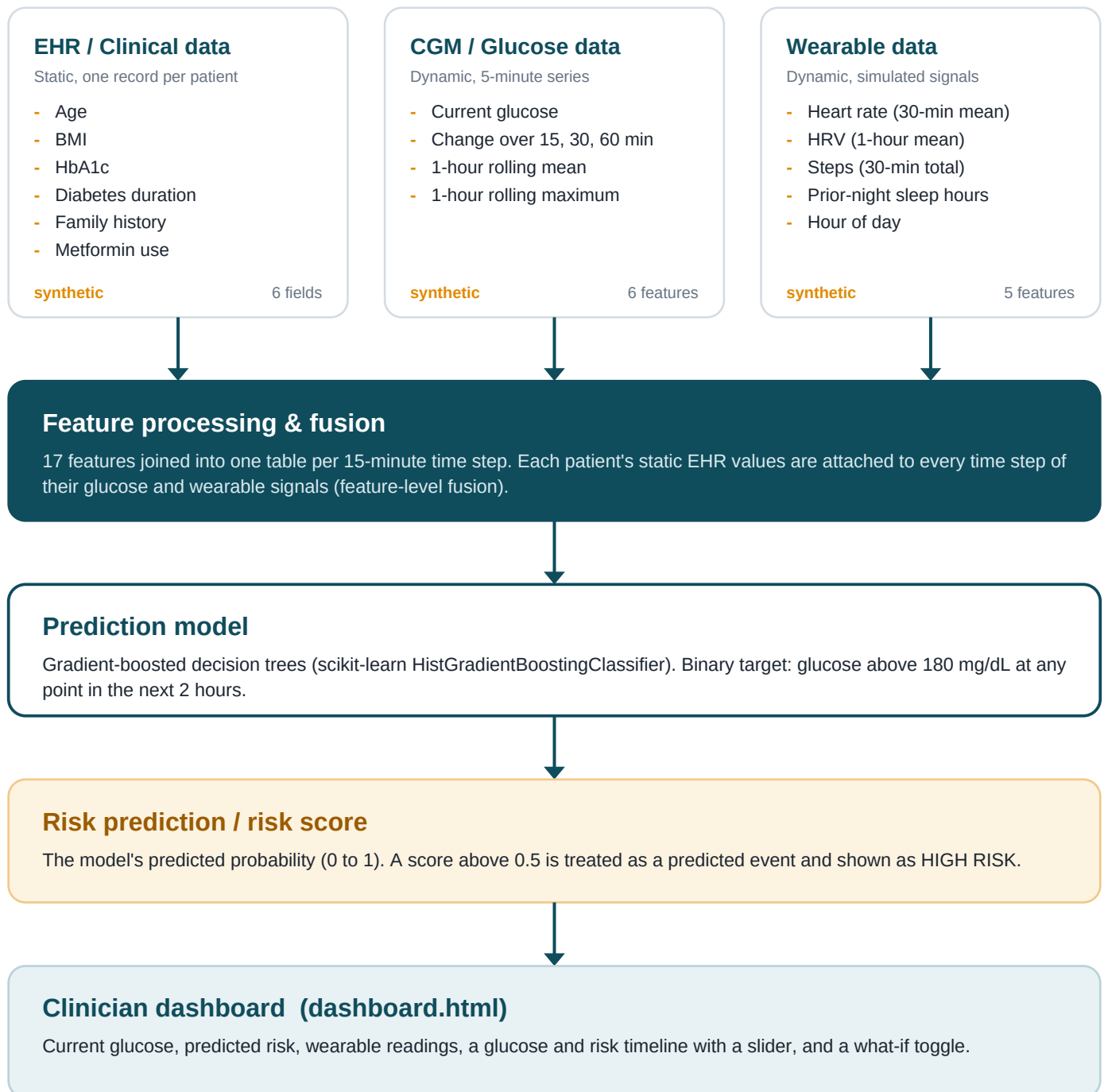


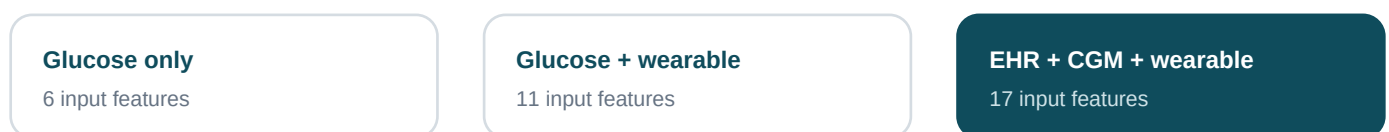
System Architecture

T2D Digital Twin | Digital Twin Challenge 2026 | Happiest Health

IMPLEMENTED PIPELINE



MODEL VARIANTS EVALUATED (same pipeline, different inputs)



Setup: 300 synthetic patients, 14 days each, 5-minute sampling; training rows kept every 15 minutes; patient-level split with 25% of patients held out for testing.

Component details

1 Data sources

All data is synthetic and generated inside **twin_poc.py** with a fixed seed. Two streams: a static EHR-style table (300 patients, 6 fields) and dynamic time series (14 days per patient at 5-minute resolution: glucose, heart rate, HRV, steps; sleep hours per night). No real patient data is used.

2 Data generation

Baseline glucose is derived from HbA1c. Three meals a day add glucose excursions, scaled by an insulin-resistance proxy built from BMI, diabetes duration and metformin use; short walks lower glucose. Sleep and stress modulate glucose. Heart rate, HRV and steps are simulated from activity, stress and age.

3 Feature engineering

Glucose: change over 15, 30 and 60 minutes; 1-hour rolling mean and maximum. **Wearable:** 30-minute mean heart rate, 1-hour mean HRV, 30-minute steps, prior-night sleep, hour of day. **Label:** 1 if glucose exceeds 180 mg/dL at any point in the next 120 minutes.

4 Feature fusion

Feature-level (early) fusion: static EHR values are attached to every time step and all 17 features go to one model together. The same pipeline is also run with fewer inputs (glucose only; glucose + wearable) to measure what each additional stream contributes.

5 Prediction model

scikit-learn **HistGradientBoostingClassifier** (gradient-boosted decision trees): 200 iterations, learning rate 0.08, fixed random seed. A binary classifier that outputs the probability of a hyperglycemic event within 2 hours.

6 Evaluation

Patient-level split (GroupShuffleSplit): 25% of patients are held out, so no patient appears in both training and test data. Test set: 75 patients, 99,900 rows, 30.6% event rate. Metrics: ROC-AUC, PR-AUC, recall and precision at a 0.5 threshold.

7 Risk output

The predicted probability is the risk score, shown as 0-100% on the dashboard. A score above 0.5 is treated as a predicted event: recall and precision are measured at this threshold and the dashboard flags HIGH RISK above it. The score has not been calibrated or validated on real patients.

8 Clinician dashboard

A static HTML/JavaScript page (no server). It replays risk scores precomputed by the model for one synthetic test patient over 24 hours (96 steps of 15 minutes), with current glucose, wearable readings and a glucose/risk timeline. The what-if walk toggle applies a fixed illustrative reduction to the displayed risk; it does not re-run the model.

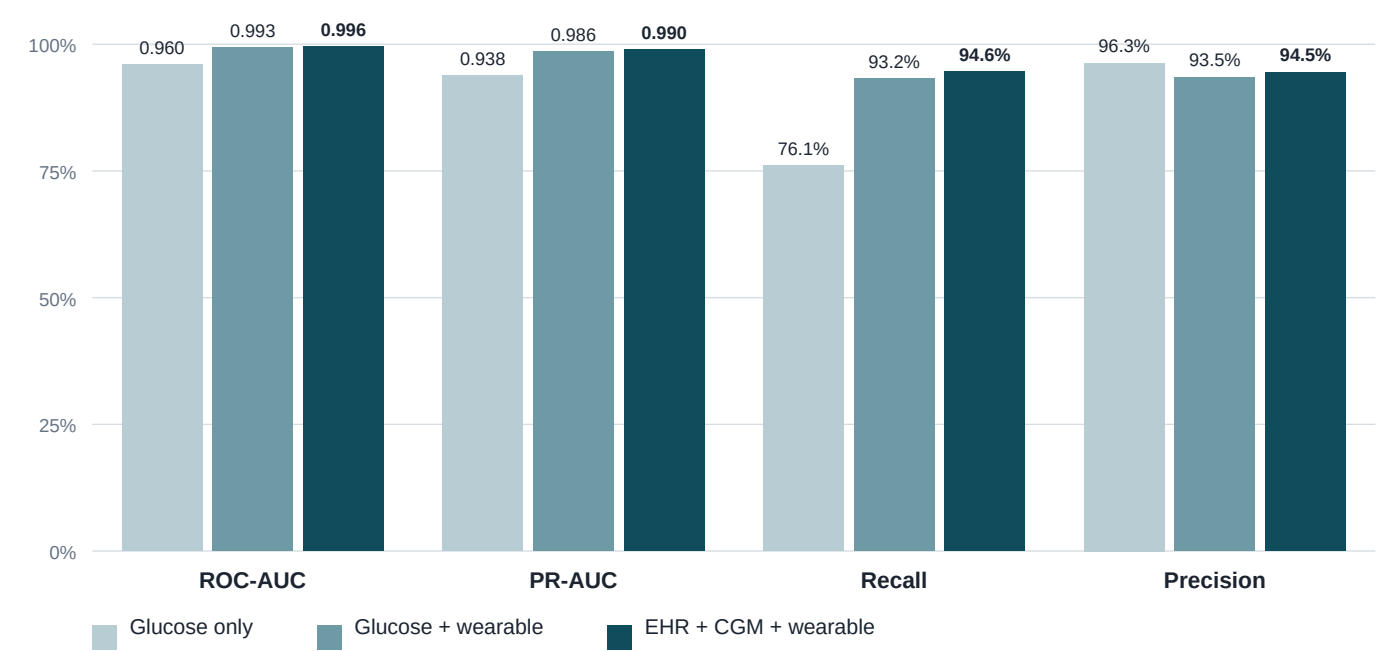
Evaluation & limitations

RESULTS ON THE HELD-OUT SYNTHETIC TEST SET

Model inputs	Features	ROC-AUC	PR-AUC	Recall	Precision
Glucose only	6	0.960	0.938	76.1%	96.3%
Glucose + wearable	11	0.993	0.986	93.2%	93.5%
EHR + CGM + wearable (fused)	17	0.996	0.990	94.6%	94.5%

Test set: 75 held-out synthetic patients, 99,900 rows, 30.6% event rate. Recall and precision use a 0.5 probability threshold.

METRIC COMPARISON



Important limitations

- **Synthetic data only.** Every record and signal is generated by the project's own simulator; no real patient data was used.
- **Not clinically validated.** No clinician review, clinical study or external validation has been performed.
- **Results reflect the generated test data.** High scores show the pipeline works end to end; they do not predict real-world performance.
- **Not medical advice.** This proof of concept must not be used for diagnosis or treatment decisions.
- **Further work needed.** Validation on real-world clinical data is required before any practical use.